



GENERAL METHOD FOR CORRECTING WAVEFORM MODELING ERROR

Ryan Johnson, Jocelyn Read

Department of Physics, California State University, Fullerton



Abstract

Gravitational-wave models for the inspiral and merger of black holes and neutron stars have intrinsic uncertainties. These may arise from different approximations of general relativity used to generate the waveform, as well as from potential modifications to general relativity. Differences between models with the same astrophysical parameters can be quantified as corrections to the frequency domain amplitude, $\Delta\mathcal{A}(f)$, and phase, $\Delta\phi(f)$. In this work, we use spline inspired phenomenological models to add variability to a baseline waveform model during parameter estimation runs. Prior distributions of the correction parameters are motivated by the faithfulness of different waveform models with each other in the signal's posterior parameter space. We demonstrate that injected waveform modifications can be recovered, and that adding waveform variability during parameter estimation runs can reduce systematic bias in recovered astrophysical parameters.

Introduction

Gravitational waves are measured by large Michelson interferometers, such as LIGO Hanford and LIGO Livingston. The waveform is described by time domain gravitational-wave strain, h , which is complex. Fourier transforming this quantity, we find the frequency domain gravitational wave, which we will use most often:

$$\tilde{h}(f) = \mathcal{A}(f)e^{i\phi(f)}. \quad (1)$$

We can approximate a gravitational wave using a gravitational-wave model, or $\tilde{h}(f) \approx \mu(f; \vartheta)$, where we let ϑ be the set of source parameters. We can equate the true gravitational wave and the model by including amplitude and phase correction terms [1]:

$$\tilde{h}(f) = \mu(f; \vartheta)(1 + \Delta\mathcal{A}(f)) \exp(i\Delta\phi(f)). \quad (2)$$

Smooth Interpolation

For our parameterization of the waveform correction functions, we choose to use 1D interpolation functions [2], which we notate as [3]:

$$f_{1D}(x; \{x_i, y_i\}), \quad (3)$$

where the points $\{x_i, y_i\}$ are the interpolation points. For our purposes, we cannot keep the sharp corners that show up at these points, so we choose to apply a moving average function to smooth them out. We then define the smooth interpolation function:

$$f_{SI}(x; \{x_i, y_i\}) = \frac{1}{2\gamma N} \int_{x-\gamma N}^{x+\gamma N} f_{1D}(x'; \{x_i, y_i\}) dx', \quad (4)$$

where γ is the smoothing parameter and N is the number of points in f_{1D} . This number depends on the system. For gravitational waves, $N = 1 + f_s t_d/2$, where f_s is the sampling frequency and t_d is the time duration.

Variable Frequency Nodes

The k^{th} frequency node is defined as

$$f_k = \frac{c^3}{GM} \xi_0 \left(1 + \left[\frac{\xi_{\text{high}} - \xi_0}{\xi_0} \right] \delta\tilde{\xi} \right)^{\frac{k}{n}}, \quad (5)$$

where n is the number of frequency nodes (excluding zero), ξ_0 is the position of the zeroth frequency node in dimensionless frequency, and $\delta\tilde{\xi}$ is a parameter that defines where the n^{th} frequency nodes is in dimensionless frequency. We convert back to frequency with the conversion factor c^3/G . ξ_0 ranges between ξ_{low} and ξ_{high} , and $\delta\tilde{\xi}$ ranges between 0 and 1. The priors for these parameters are based on where we expect the gravitational-wave amplitude to be above noise in the interferometers.

ξ_{low} and ξ_{high} are the absolute boundaries on dimensionless frequency ξ , which is defined as $\xi = GMf/c^3$, where M is the total system mass, $m_1 + m_2$, c is the speed of light, and G is Newton's gravitational constant.

General Correction Model

$$\mu_{GC}(f; \vartheta, A, \Phi) = \mu(f; \vartheta)(1 + \Delta\mathcal{A}_{SI}(f; \{f_k, \tilde{\alpha}_k \sigma_{\alpha,k}\})) \exp(i\Delta\phi_{SI}(f; \{f_k, \tilde{\varphi}_k \sigma_{\varphi,k}\}))$$

$$A = \{\tilde{\alpha}_0, \tilde{\alpha}_1, \dots, \tilde{\alpha}_n, \xi_0, \delta\tilde{\xi}, M, \gamma\}$$

$$\Phi = \{\tilde{\varphi}_0, \tilde{\varphi}_1, \dots, \tilde{\varphi}_n, \xi_0, \delta\tilde{\xi}, M, \gamma\}$$

Waveform Correction Parameters

We assign the variable α to define the amplitude correction and φ to define the phase correction. The prior distributions for these parameters are defined by draws of waveform model differences, which are amplitude and phase differences between two waveform models with the same source parameters. The priors we obtain from this depend on the locations of the frequency nodes. Because prior distributions must be constant, we define new parameters:

$$\tilde{\alpha}_k = \alpha_k / \sigma_{\alpha,k}, \quad \sigma_{\alpha,k} = \sqrt{(\overline{\Delta\mathcal{A}_\mu}(\xi_k))^2 + (\delta\mathcal{A}_\mu(\xi_k))^2}, \quad (6)$$

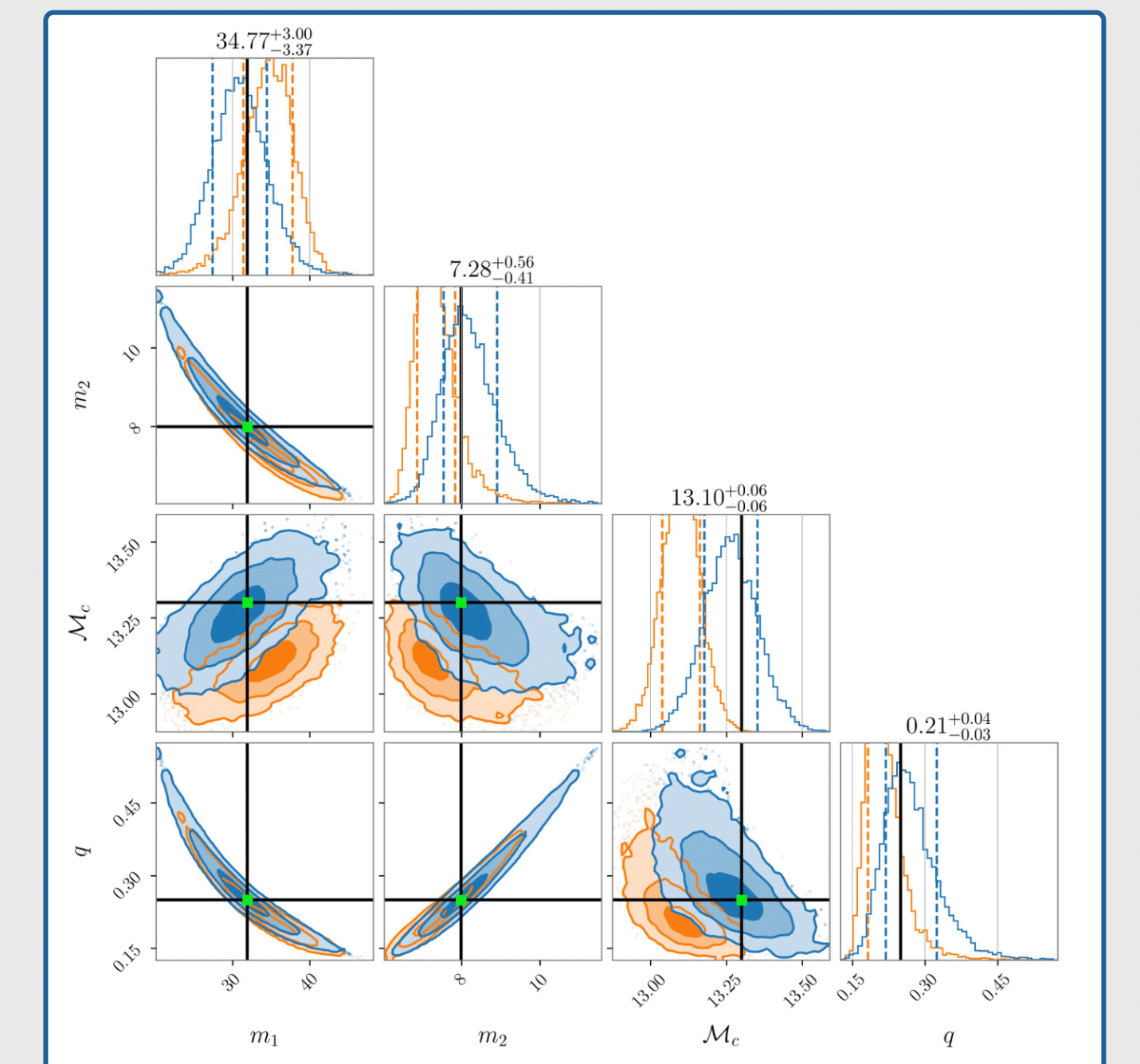
$$\tilde{\varphi}_k = \varphi_k / \sigma_{\varphi,k}, \quad \sigma_{\varphi,k} = \sqrt{(\overline{\Delta\phi_\mu}(\xi_k))^2 + (\delta\phi_\mu(\xi_k))^2}, \quad (7)$$

where the overlined quantities represent the means of the waveform model difference draws and the δ quantities are their corresponding standard deviations. We also choose to evaluate these priors in dimensionless frequency units. Using $\tilde{\alpha}$ as an example, we see that if we define the prior on $\tilde{\alpha}_k$ to be $\mathcal{N}(0, 1)$, then the corresponding α_k will be equivalent to taking a draw from $\mathcal{N}(0, \sigma_{\alpha,k})$. The same is true for $\tilde{\varphi}$. This effectively gives us variable priors.

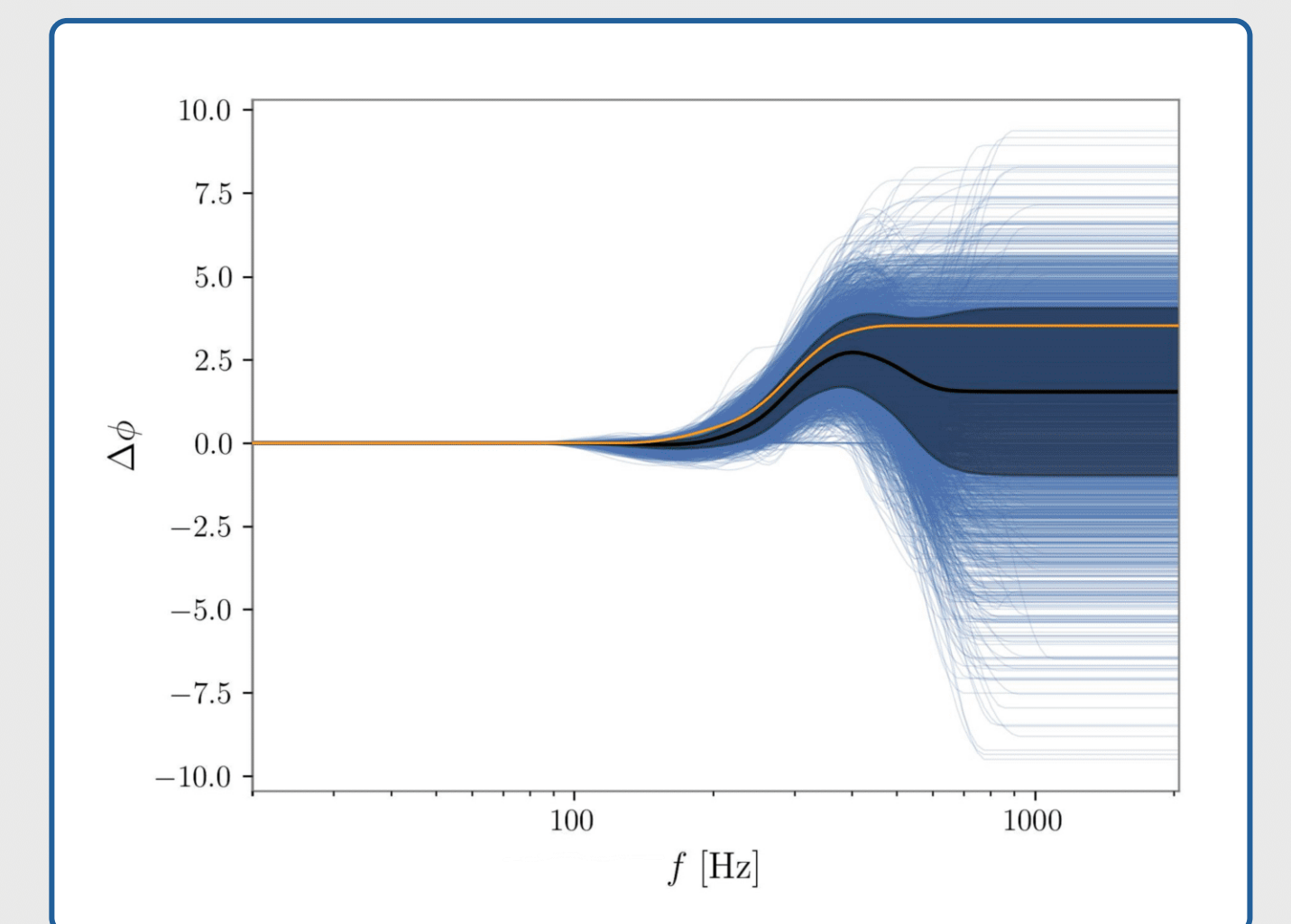
Results

In this test, we inject an artificial phase error to observe how both a default waveform model [4] and our corrected waveform model behave. We expect that the presence of a phase error will cause deviations in the output of the default waveform model. Our model should be able to account for the phase error and return a more accurate result.

On the right, we see the posteriors for the mass parameters of a binary black hole test [5]. In orange, we show the result from a default waveform model. In blue, we show the result from our corrected model. In black and lime, we show the true values of these parameters. We can see that our model recovered more accurate values for these parameters.



Here (right), we show the phase correction posterior in blue and black. In orange, we show the injected artificial phase error. We see that the injected phase error was recovered well. We also observe that the recovered phase shifts to return the prior after $f \approx 600$ Hz due to the shrinking amplitude of the signal.



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